COURSE-BASED PROJECT DOCUMENTATION | DOC-16

Test Cases & Final Verified Results

Breakdown of 59 Parser Tests & 14 GUI Integration Tests (100% Pass Rate)

1. Final Verified Test Summary

VERIFIED TEST RESULTS SUMMARY
PARSER TEST SUITE: 59 / 59 PASSED (100.0%) GUI INTEGRATION SUITE: 14 / 14 PASSED (100.0%) TOTAL AUTOMATED TESTS: 73 / 73 PASSED (100.0%)

2. Breakdown of 59 Parser Tests by Category

Test Category	Test Count	Pass Rate	What the Category Proves
Original Regression	23 Tests	23 / 23 (100%)	Core Java 8?17 classes, methods,...
Java 21 Feature Tests	12 Tests	12 / 12 (100%)	Record patterns, switch patterns...
Phase 9 Completeness	8 Tests	8 / 8 (100%)	>>>= assignment, module-info, mu...
Error Accuracy (Phase 10)	13 Tests	13 / 13 (100%)	Exact line accuracy after blanks...
Stress & Boundaries	3 Tests	3 / 3 (100%)	Deep nested record deconstructio...
Total Parser Tests	59 Tests	59 / 59 (100%)	Complete front-end grammar and e...

3. Breakdown of 14 GUI Integration Tests

All 14 integration tests in tests/test_gui_backend.py passed with 100% success:

- ? 1. test_window_title_and_structure (Window and subwidgets)
- ? 2. test_backend_initialization (ctypes CDLL handle)
- ? 3. test_valid_simple_class (In-memory simple valid check)
- ? 4. test_valid_java21_features (In-memory Java 21 records & switch)
- ? 5. test_invalid_line_1 (Error on line 1 extraction)
- ? 6. test_invalid_middle_line (Error on line 4 in method body)
- ? 7. test_invalid_after_comments_and_blank_lines (Line 6 error)
- ? 8. test_invalid_eof (Missing closing brace at EOF)
- ? 9. test_gui_file_load_and_line_numbers (File load & gutter sync)
- ? 10. test_gui_editor_editing_and_rechecking (Invalid -> Fix -> Valid workflow)
- ? 11. test_gui_error_highlight_and_clearing (Tag add and clear on edit)
- ? 12. test_gui_long_file_scrolling_and_line_numbers (100+ line gutter alignment)
- ? 13. test_command_palette_filtering (Fuzzy search & execution)
- ? 14. test_theme_switching (Dark/light theme toggle)